

Archel Taneka Sutanto

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SUMMARY

Data Scientist with 3+ years of industry experience applying machine learning, predictive modelling and controlled experimentation to customer journeys and commercial outcomes. Delivered measurable booking conversion and personalisation gains using behavioural data and recommendation systems. Skilled in Python and SQL, with independent project experience in agentic AI, retrieval-augmented generation (RAG) and causal inference.

WORK EXPERIENCE

tiket.com

Jakarta, Indonesia

Associate Data Scientist

Oct 2022 – Jul 2024

Online travel platform serving millions of customers across flights, hotels, rail and attractions.

- Generated approximately GBP 3.97M (IDR 94B) incremental gross booking value (GBV) through a scikit-learn payment recommendation system analysing customer behaviour across 15+ payment methods, validated through controlled A/B experimentation.
- Identified high-value personalisation opportunities through CatBoost-based recommendation-system experimentation and segmented traveller behaviour analysis, delivering approximately GBP 338K (IDR 8B) incremental GBV.
- Increased hotel booking conversion by 2.3%, generating approximately GBP 1.99M (IDR 47B) annual GBV, by deploying a YOLOv10 computer vision model with TensorRT inference to identify low-quality images across 15,000+ hotel listings.
- Saved approximately GBP 13.3K (IDR 315M) annually by deploying a Keras/TensorFlow email prediction system to extract flight details from customer reschedule emails.

Sayurbox

Jakarta, Indonesia

Junior Data Scientist

May 2021 – Oct 2022

Jakarta-based e-grocery and supply chain platform focused on fresh produce and last-mile fulfilment.

- Developed and maintained weekly demand forecasts for hundreds of SKUs, supporting inventory planning through recurring reports and continuous model performance monitoring.
- Automated warehouse staff allocation through the Picker Checker system, reducing coordination time from hours to minutes and improving workforce planning across multiple warehouse locations.

PROJECTS

ExperimentOS AI

July 2026 – Sep 2026

Independent Research

- Designed and developed an agentic RAG platform using FastAPI, LangGraph, PostgreSQL, and pgvector, querying over product experimentation data through semantic retrieval, multi-agent orchestration, and grounded LLM responses with citation support.
- Built automated RAG evaluation and prompt regression testing with RAGAS and DeepEval, supported by structured outputs, AI observability, Dockerized development and GitHub Actions CI/CD.

Causal Inference & Uplift Modeling Platform

Independent Research

- Built an end-to-end experimentation and causal inference framework comparing Propensity Score Matching, Double Machine Learning, and uplift modelling approaches for estimating heterogeneous treatment effects from observational data.
- Developed reproducible analytical pipelines using Python, EconML, and CausalML to estimate ATE and CATE, enabling comparison of treatment effectiveness across customer segments rather than relying solely on average outcomes.
- Created an interactive Streamlit application that translated complex causal inference outputs into business recommendations suitable for product and commercial stakeholders.

EDUCATION

Monash University

Melbourne, Australia

M.Sc. Data Science

Aug 2024 – Oct 2026 (expected graduation)

University of Nottingham

Nottingham, United Kingdom

B.Sc. (Hons.) Computer Science — First-Class Honours

Sep 2019 – Jul 2020

Dual degree with Bina Nusantara University.

Bina Nusantara International

Jakarta, Indonesia

B.Sc. Computer Science — GPA: 3.74/4.00

Sep 2016 – Sep 2020

TECHNICAL SKILLS

Programming: Python, SQL, R

Analytics: A/B Testing, Experimentation, Causal Inference, Uplift Modelling, Forecasting

Machine Learning: scikit-learn, PyTorch, TensorFlow, CatBoost, EconML, CausalML

AI Development: LLMs, RAG, Agentic AI, LangGraph, RAGAS, DeepEval (independent projects)

Data Platforms: GCP, BigQuery, PySpark, Databricks, PostgreSQL, pgvector

Engineering: FastAPI, Docker, Git, GitHub Actions